

Quarto for Scientific Writing

Reproducible documents for students & researchers

Workshop · Zoom

30 June 2026

Section 1

Why Quarto?

Who is this workshop for?

- **Students** — quantitative thesis: data, tables, figures, citations, campus-formatted PDF.
- **Lecturers & researchers** — **reproducible** papers, reports, and presentations.

One tool, one workflow — from analysis to finished document.

The old way hurts

Word + SPSS/Excel	With Quarto
Copy numbers by hand → typos	Numbers automatic from code
Table/figure numbering breaks	Numbering & TOC automatic
Revise data = redo & re-paste	Change data → re-render
Manual citations	Automatic citations (APA, Zotero)
Slides built separately	Slides from the same data

What is Quarto?

A publishing system: **prose + code + data** in one .qmd file.

.qmd Quarto PDF · HTML · Word · Slides

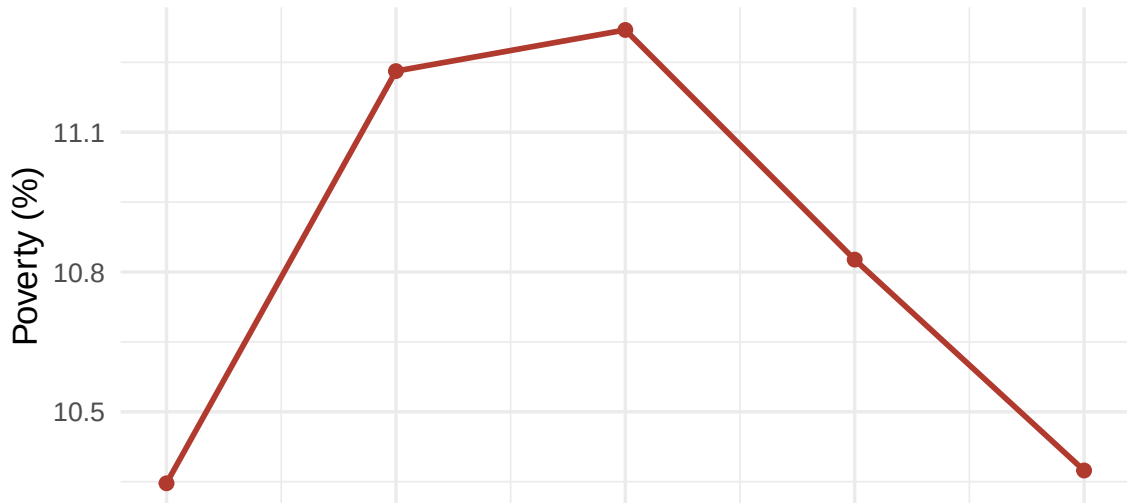
- Free & open source. Supports **R** and **Python**.
- **Write once, render many.**

Section 2

The core: data lives in the document

Numbers & charts straight from data

The mean poverty rate in Central Java **2023** = **10.37%** (computed at render time, not typed).



Maps too, from the same data

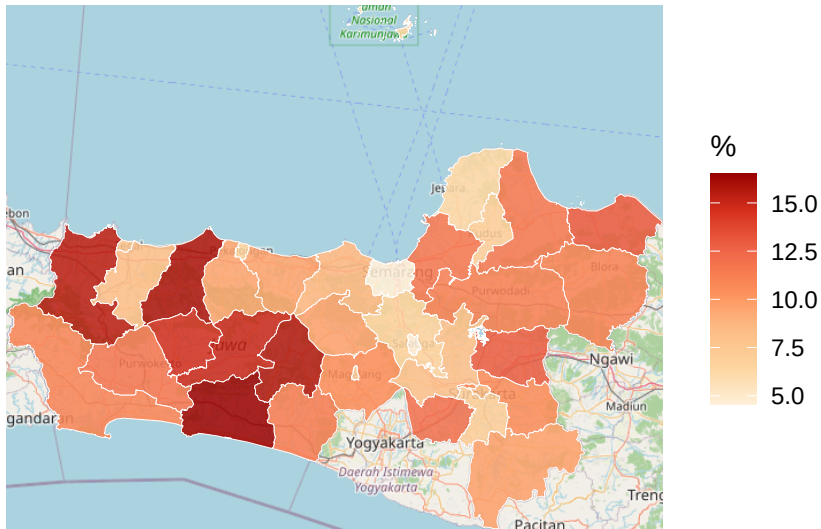


Figure 2: Poverty by district, 2023

The killer feature: *reproducible*

Change one number in the data → the whole document updates: inline numbers, tables, figures, **maps**, even **slides**.

No more manual copy-paste. Advisor asks to change the year range? Re-render once.

One gotcha

`quarto render` — not just the *preview* button.

- The live preview **does not watch data files**.
- After editing **data**, run `quarto render` and open the result.

Section 3

An analysis walkthrough

Descriptive statistics

Variable	Mean	Median	SD
kemiskinan	10.37	10.61	3.23
pdrb_kapita	41.88	41.05	5.66
inflasi	2.69	2.67	0.46
ipm	79.39	79.17	1.92

Central tendency (mean, median) and spread (SD) for each variable — 2023.

Plotting: relationships

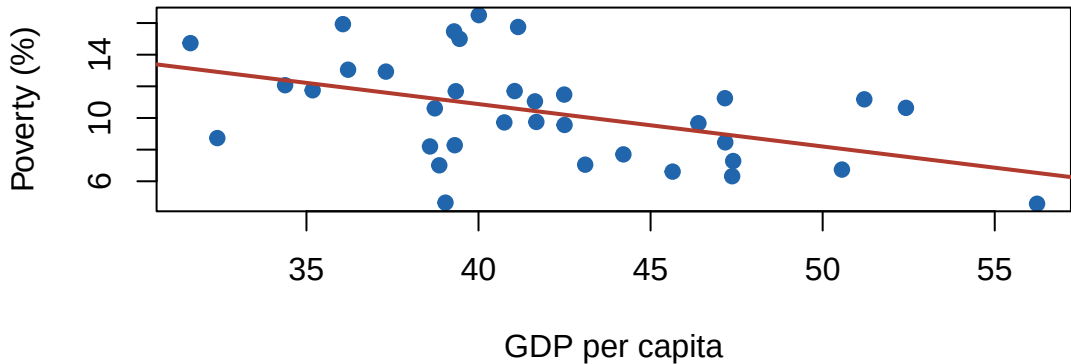


Figure 3: Poverty vs GDP per capita (2023)

Multiple linear regression

	Coefficient	p-value
Intercept	98.355	0.000
GDP/capita	-0.123	0.001
Inflation	0.198	0.112
HDI	-1.077	0.000
Unemployment	0.235	0.031

**** $R^2 = 0.533$. Read the coefficient sign (direction) & p-value** (significance).** Low $R^2 \Rightarrow$ needs fixed effects / spatial regression.

Does poverty **cluster** spatially? Test with **Moran's I**:

Moran's I = 0.294 ($p = 0.0038$). Positive & significant $\Rightarrow$ poverty “**pockets**” (spatial clusters). Next: LISA & spatial regression.

Section 4

For your work

Students: a pre-formatted thesis

- **Chapter I–V** template, cover page, approval page — to campus guidelines.
- “BAB I” numbering, **Table 4.1 / Figure 4.2**, TOC/lists/bibliography — automatic.
- Write the content, swap in your data, hit **Render** → thesis PDF.

- **Papers & reports:** one source → PDF/Word/HTML; citations via **Zotero**.
- **Presentations:** **Reveal.js** (interactive HTML) & **Beamer** (PDF) from one file.
- Collaborate via **Git**; publish to **GitHub Pages** automatically.

One source, two kinds of slides

```
format:
  revealjs: default      # HTML slides (screen/web)
  beamer:                # PDF slides (projector)
  aspectratio: 169
```

```
quarto render slide.qmd --to revealjs  # -> slide.html
quarto render slide.qmd --to beamer    # -> slide.pdf
```

These very slides are made that way — Beamer for PDF, Reveal.js for the web.

Section 5

Get started now

How to start

- 1 Install **R + RStudio** (Quarto is included).
- 2 `quarto install tinytex` (for PDF).
- 3 (*Optional*) Python + `reticulate` for Python examples.
- 4 Create `hello.qmd`, hit **Render**. Done.

- **Hands-on tutorial (1 hour):** this workshop site (*Hands-on* menu).
- **Book “Quarto untuk Skripsi” (Indonesian):**
<https://firmanhadi21.github.io/thesis-with-quarto/>
- **Quarto docs:** <https://quarto.org>

Thank You

Let's write reproducibly.